Ziwei Liu



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Ph.D. student in the AML Lab at the School of Data Science, City University of Hong Kong. My research focuses on Recommender Systems, Information Retrieval, and Large Language Models, with papers published or preprinted at top international AI conferences.

EDUCATION

City University of Hong Kong 2025.09 - Present
Ph.D. in Data Science. Affiliation: AML Lab

City University of Hong Kong 2023.09 - 2024.10
M.E. in Data Science. Supervisor: Prof. Zhao Xiangyu

Southeast University 2019.09 - 2023.06
B.E. in Robotics Engineering. Co-supervised by Prof. Gan Yahui and Prof. Li Jun; Excellent Graduation Project

RESEARCH EXPERIENCE

Chinese University of Hong Kong, Shenzhen 2024.05 - 2024.12
Research Assistant

PUBLICATIONS & PREPRINTS

Conditional Memory Enhanced Item Representation for Generative Recommendation (ComeIR)

Ziwei Liu*, Yejing Wang*, Shengyu Zhou, Xinhang Li, Xiangyu Zhao
Preprint, 2026

- ComeIR uses token scoring and dual-level Engram memories to preserve item identity and SID structure, then reuses the memories during decoding to bridge item-level inputs and token-level generation.

The Best of Both Worlds: Harmonizing Semantic and Hash IDs for Sequential Recommendation (H²Rec)

Ziwei Liu*, Yejing Wang*, Wanyu Wang, Zejian Wang, Qidong Liu, Zijian Zhang, Wei Huang, Chong Chen, Xiangyu Zhao
ACM SIGKDD Conference on Knowledge Discovery and Data Mining, ADS Track (KDD), 2026 · CCF-A

- H²Rec combines the semantic generalization of SIDs with the collaborative uniqueness of HIDs, balancing recommendation quality for both head and long-tail items.

LLM-EDT: Large Language Models Enhanced Cross-domain Sequential Recommendation with Dual-phase Training

Ziwei Liu*, Qidong Liu*, Wanyu Wang, Yejing Wang, Pengyue Jia, Tong Xu, Wei Huang, Chong Chen, Xiangyu Zhao
ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2026 · CCF-A

- LLM-EDT addresses domain imbalance, noisy augmentation, and rough profiling via transferable item augmentation, domain-specific fine-tuning, and adaptive domain-aware preference aggregation.

SIGMA: Selective Gated Mamba for Sequential Recommendation

Ziwei Liu*, Qidong Liu*, Yejing Wang, Wanyu Wang, Pengyue Jia, Maolin Wang, Zitao Liu, Yi Chang, Xiangyu Zhao
AAAI Conference on Artificial Intelligence (AAAI), 2025 · CCF-A

- SIGMA equips Mamba with partially flipped bidirectional modeling, an input-dependent selective gate, and a feature-extracting GRU for efficient long- and short-term sequential recommendation.

TUTORIALS

- **KDD 2026 (CCF-A):** Tutorial on Generative Recommendation: Foundations and Frontiers. Project page. 2026.08.
- **KDD 2025 (CCF-A):** Large Language Model Enhanced Recommender Systems: Taxonomy, Trend, Applications, and Future Directions. Project page. 2025.08.

HONORS AND AWARDS

- KDD 2026 Student Travel Award and Volunteer, 2026.06.
- AAAI 2025 Travel Award and Volunteer, 2025.01.
- Silver Medal, Kaggle LLM Prompt Recovery (44/2,175), 2024.12.
- Excellent Graduation Project, Southeast University, 2023.06.

ACADEMIC SERVICE

Conferences

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2027), ADS Track
- AAAI Conference on Artificial Intelligence (AAAI 2027), Main Track
- ACM International Conference on Multimedia (MM 2026), Main and DB Tracks
- ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2026)
- AAAI Conference on Artificial Intelligence (AAAI 2026), Main and AIA Tracks
- ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2025)
- ACM Conference on Recommender Systems (RecSys 2025)

Journals

- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- ACM Transactions on Knowledge Discovery from Data (TKDD)
- ACM Transactions on Information Systems (TOIS)